Evaluating the Performance of LLMs in Drafting Data Management Plans

FAIR DATA INNOVATIONS HUB

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DMP Tool — Making best practice easier

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Background

Most funders now require Data Management Plans (DMPs) to be submitted as part of a grant proposal. A DMP is a document that describes how researchers will manage, preserve, and share data from their projects in line with the Findable, Accessible, Interoperable, and Reusable (FAIR) principles and other relevant disciplinary guidelines. Its aim is to ensure researchers plan in advance for effective data management and sharing.

Preparing a DMP can be difficult for researchers because they often lack training and knowledge in data management practices. Librarians are available to assist at larger institutions, but they face similar challenges and do not always have the resources to assist with all aspects of a DMP.

Open-source platforms like DMPTool provide funder guidance, sample DMPs, and templates, but researchers and librarians still need to manually complete sections that vary based on domain and funder requirements.

Purpose

Given the rise of Large Language Models (LLMs) being used to write technical documents, we evaluated an open-source LLM, Llama 3.3, and a commercial LLM, GPT 4, in drafting NIH DMPs compliant with the National Institutes of Health (NIH)'s 2023 guidelines.

The goal was to investigate whether LLMs were ready, off-the-shelf, to assist biomedical researchers in preparing their DMPs.

Methods

- Collected 26 NIH-provided human-written DMP examples from previously funded biomedical proposals. Each DMP followed the 2023 NIH DMS Policy structure, consisting of 12 sections. We used them as reference DMPs.
- Generated DMPs using GPT-4.1 and Llama 3.3 70B with a standardized minimal prompt that included the NIH Institute/Center, study details, participant information, data types, and the full NIH DMP template.
- Evaluated generated DMPs against reference NIH DMPs using ROUGE and SBERT similarity at the document, element, and sub-element levels.
- Conducted a human expert evaluation using a custom survey.
- Experts blindly reviewed three DMP versions: human-written, GPT-generated, and Llama-generated. They rated section-wise satisfaction, identified error types, provided overall satisfactions, and guessed whether each DMP was human-written or LLM-generated.

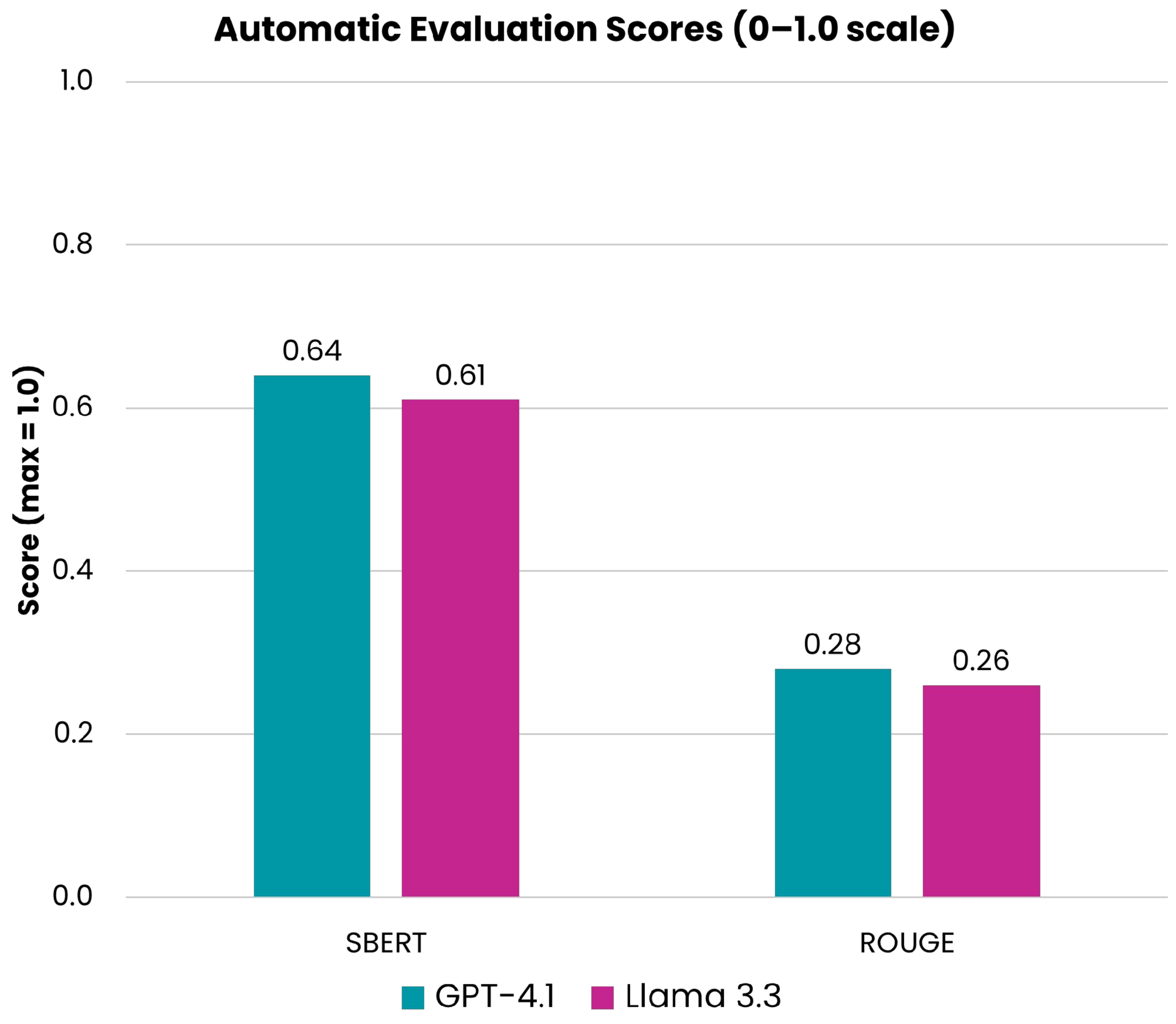


Figure 1. Overall SBERT and ROUGE in Automatic Evaluation

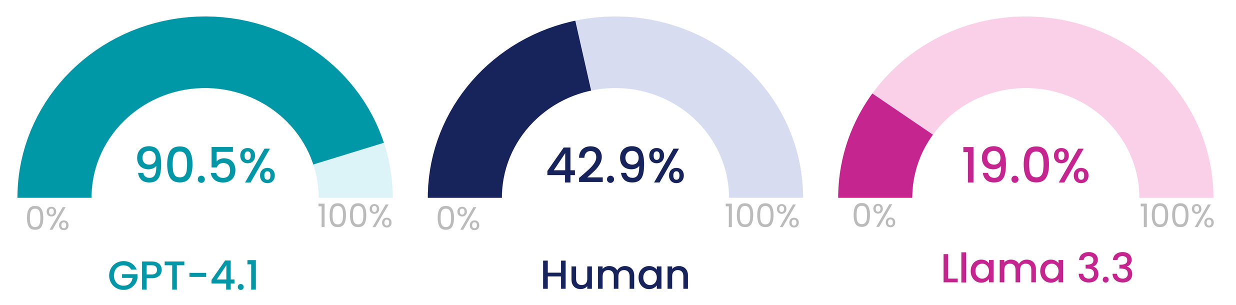


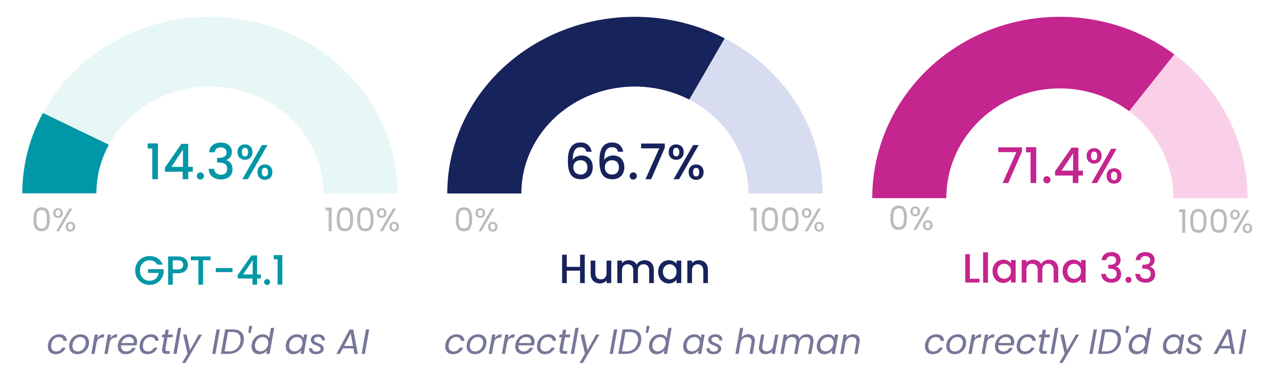
Figure 2. Satisfied or Very Satisfied (%) in Human Evaluation

GPT-4.1	Human	Llama 3.3
2.3	7.0	7.3

Top error types (errors per DMP)

GPT-4.1	• Clarity 0.42	• Completeness 0.23
Human	• Clarity 1.71	• Completeness 1.52
Llama	• Clarity 2.52	• Completeness 1.61

Figure 3. Average Errors per DMP in Human Evaluation



GPT outputs mistaken for human-written **85.7%** of the time

Figure 4. Human vs. AI Detection (% correct) in Human Evaluation

Results

The automatic reference-based evaluation scores for both ROUGE and SBERT were slightly higher for GPT than for Llama (Figure 1).

GPT achieved higher similarity to NIH references than Llama in sections tied to repository selection, access conditions, reuse considerations, and oversight.

A total of 21 human experts participated in the evaluation. Participants were primarily from Library and Information Science (54.5%) and the Biological and Life Science (31.7%).

On a 5-point Likert satisfaction scale, experts reported higher overall satisfaction with GPT-generated DMPs compared to Llama and even human-generated ones (Figure 2).

Overall, GPT showed the lowest error rates across error types in the assessment, but some clarity and completeness errors were still pointed out by participants (Figure 3).

GPT outputs were most often mistaken for human-written (Figure 4).

Conclusion

- This study establishes a template-agnostic evaluation framework and initial benchmark for future LLM–DMP research.
- We observed that the GPT model was rated slightly better in the automated assessment but outperformed Llama largely in the human assessment.
- This suggests that open-source LLMs may not be as ready off-the-shelf as commercial ones for drafting funder-compliant DMPs.
- Some errors were still reported with GPT-generated DMPs suggesting that LLM-generated DMPs may still require expert review to ensure accuracy, completeness, and compliance with funder requirements.

Future work

Our future work will extend this evaluation to other major funders, including NSF, and to the updated NIH guidelines, to assess whether the findings generalize across different DMP formats and requirements.

We will also investigate how enhancement approaches, such as retrieval-augmented generation (RAG), can improve the performance of open-source LLMs, including Llama, to make high-quality LLM-assisted DMP drafting accessible to research institutions that cannot rely on commercial APIs.

Currently, we are developing tools to automatically generate machine actionable DMPs so that LLM-based DMP-generation and evaluation could be achieved more consistently across different DMP formats.

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